

CS 489/698: Introduction to Natural Language Processing

Lecture 6.1: Language Modeling I

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Announcements

- Assignment 1 should be submitted via LEARN to be properly graded.
 - Added one DP encoder that is more suitable with SentencePiece-like algorithms.
 - Your final mark will be based on the better compression ratio applying to both algorithms.
- CS698: Two projects with different dues
 - Project 1 (shared with CS489, 25%): The Interstellar Autocomplete Challenge
 - Initial proposal due: Feb 2nd
 - Midterm check-in due: Feb 23rd
 - Final submission: Final exam weeks
 - Details at <https://github.com/waterloo-nlp/intro-to-nlp-project>
 - Project 2 (NLP-related research project at your own choice; 25%)
 - Proposal/midterm check-in due Feb 23rd: 1-page PDF
 - Final submission: Final exam weeks

Reminder

- Kaggle public leaderboard (training set) performance does (usually positively) correlate with but not guarantee the performance on held-out data.
 - Example (sentiment classification):
 - if sentence in training set:
 - return label of sentence in training set
 - else:
 - return random label in {0, 1}
- Don't optimize too hard on the training set, otherwise your model might overfit.

Suggestion: reserve some training examples to estimate the performance on the held-out data.

Outline: Language Modeling

- Language modeling
 - General approach
 - N-gram language models
 - Autoregressive language models
 - Masked language models

Recap: Distributional Semantics

You shall know a word by the company it keeps.

-- J. R. Firth, 1957

A bottle of *tezgüino* is on the table.

Everybody likes *tezgüino*.

Don't have *tezgüino* before you drive.

We make *tezgüino* out of corn.

A bottle of _____ is on the table.

Everybody likes _____.

Don't have _____ before you drive.

We make _____ out of corn.

The Shannon Game

- The Shannon game [Shannon 1951]:
How well can you predict the next letter?

(1) THE ROOM WAS NOT VERY LIGHT A SMALL OBLONG
(2) ----ROO-----NOT-V-----I-----SM----OBL-----
(1) READING LAMP ON THE DESK SHED GLOW ON
(2) REA-----O-----D-----SHED-GLO--O--
(1) POLISHED WOOD BUT LESS ON THE SHABBY RED CARPET
(2) P-L-S-----O---BU--L-S--O-----SH-----RE--C-----

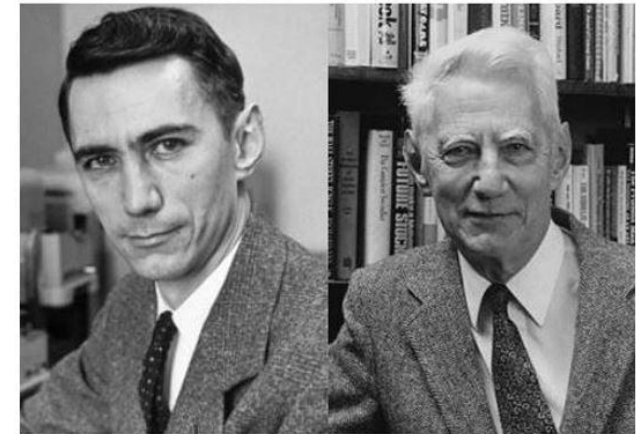
Prediction and Entropy of Printed English

By C. E. SHANNON

(Manuscript Received Sept. 15, 1950)

A new method of estimating the entropy and redundancy of a language is described. This method exploits the knowledge of the language statistics possessed by those who speak the language, and depends on experimental results in prediction of the next letter when the preceding text is known. Results of experiments in prediction are given, and some properties of an ideal predictor are developed.

Claude Shannon



30 Apr 1916 – 24 Feb 2001

Language Models: General Formulation

Language model computes a probability distribution over strings in a language:

$$P(\mathbf{x})$$

$\mathbf{x}$: A string with tokens $x_1, x_2, \dots, x_n$.

$$p(\text{'I'm not a cat'}) = 0.00000004$$

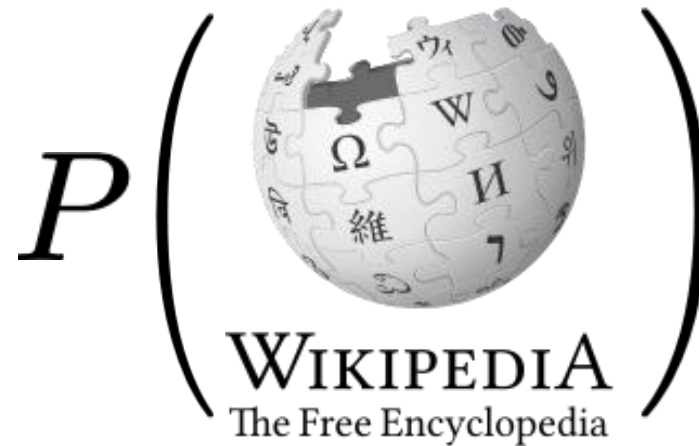
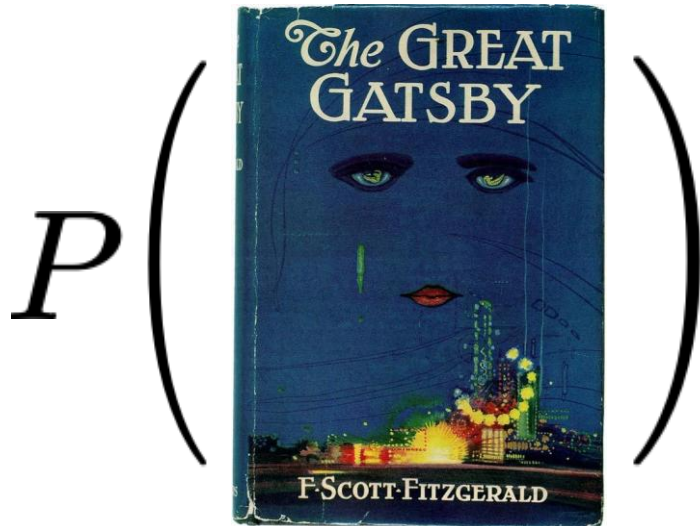
$$p(\text{He is hungry}) = 0.000025$$

$$p(\text{Dog the asd@sdf 1124 !?}) \approx 0$$

Language Models: General Formulation

Language model computes a probability distribution over strings in [a language](#):

$$P(\mathbf{x})$$



Nowadays: large language models

$$P(x)$$

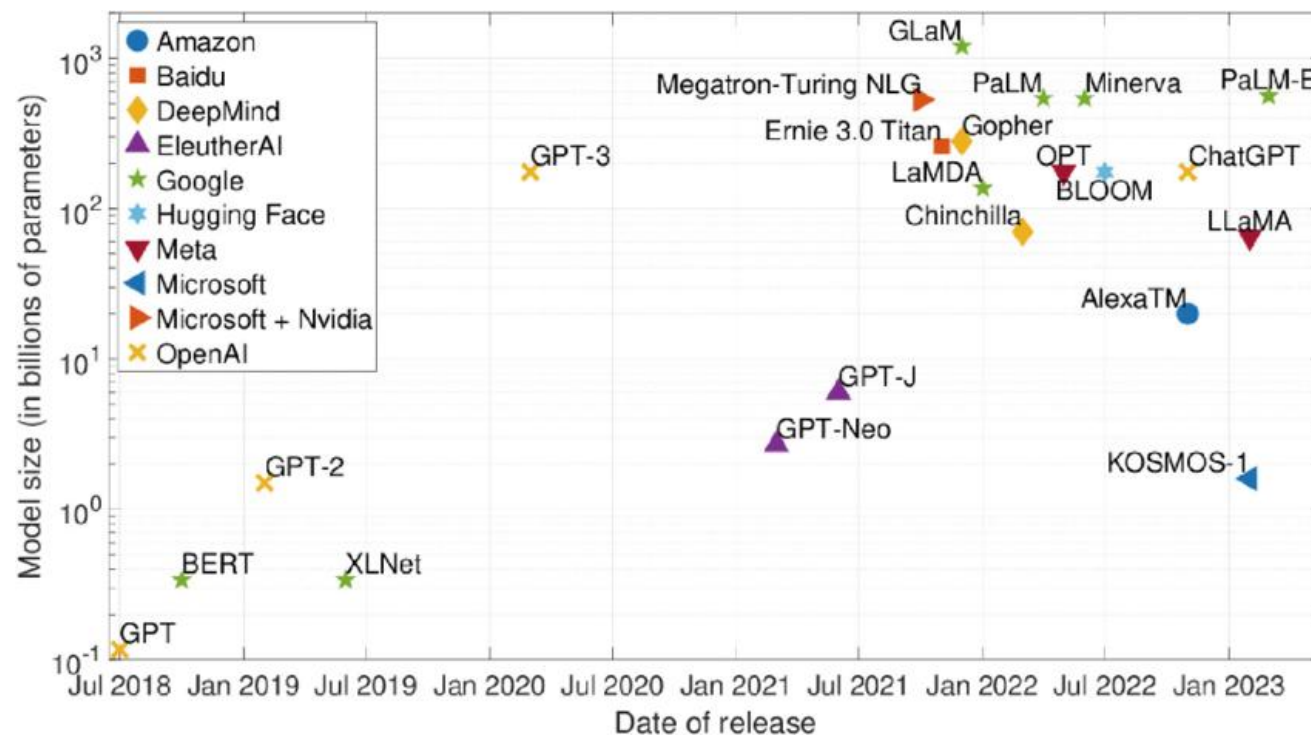
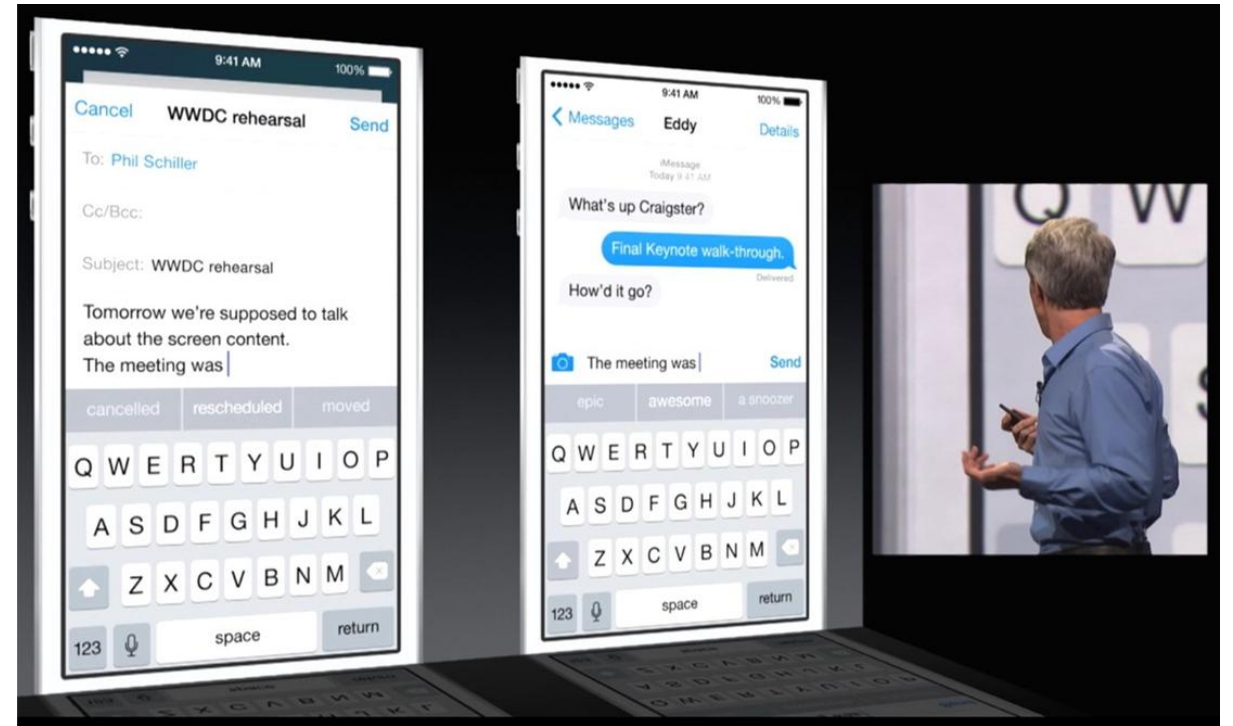
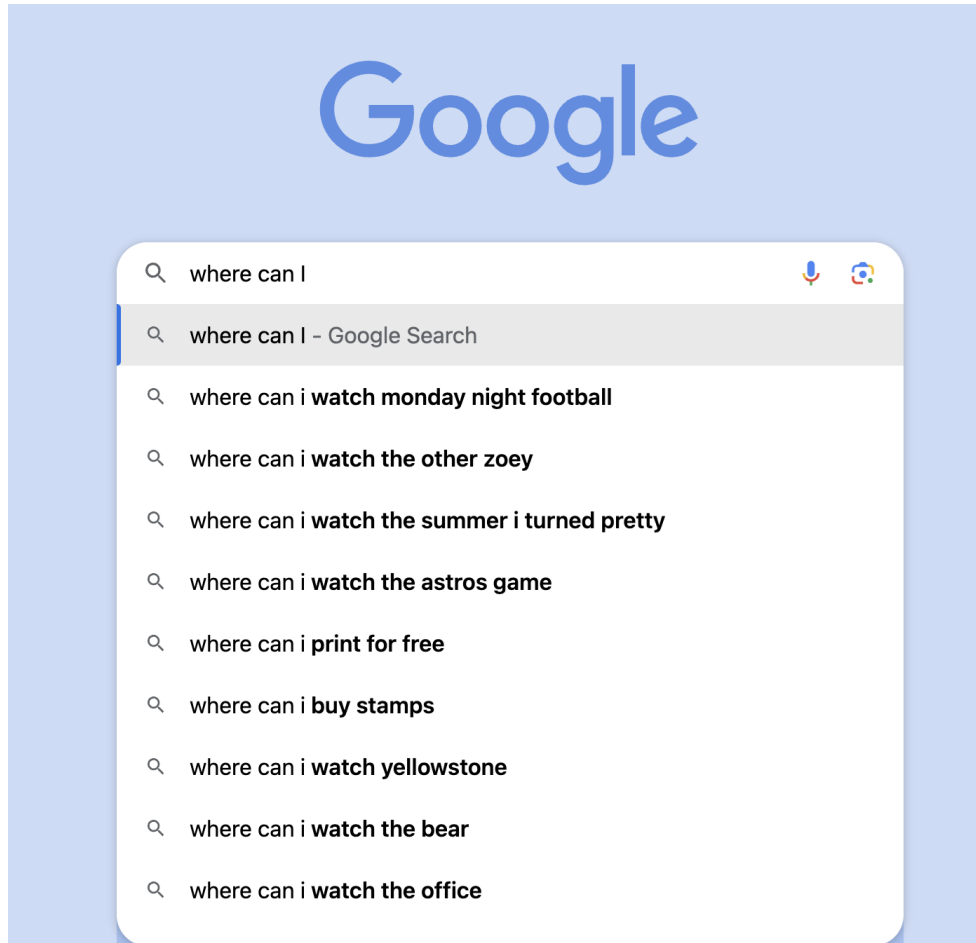


Figure credit: Kath et al. (2023)

Languages models are now everywhere!



Language Models: General Formulation

Language modeling: assign probabilities to token sequences.

Why?

- Machine translation:
 - $P(\textit{turn the camera off}) > P(\textit{put the camera out})$
- Speech recognition:
 - $P(\textit{I need to recognize speech}) > P(\textit{I need to wreck a nice speech})$
- Spelling/grammar correction:
 - The office is about fifteen minuets from my house
 - $P(\textit{about fifteen minutes from}) > P(\textit{about fifteen minuets from})$
- (modeling) Define a statistical model $P_{\theta}(\mathbf{x})$, where $\mathbf{x}$ is a string.
- (learning) Estimate the parameters θ from data (by maximizing likelihood).

Language Modeling

Goal – Compute the probability of a sequence of words:

$$P(\mathbf{x}_{1:n}) = P(x_1, x_2, \dots, x_n)$$

Relatedly, modeling probability of the next word:

$$P(x_n \mid x_1, x_2, \dots, x_{n-1})$$

Relatedly, modeling the probability of a masked word given its context:

$$P(x_k \mid x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n)$$

A model that computes any of the above is called a **language model**.

Language Modeling

$$P(\mathbf{x}_{1:n}) = P(x_1, x_2, \dots, x_n)$$

How do we model this?

Recap: the chain rule of probability.

$$P(A, B) = P(A)P(B \mid A)$$

$$\begin{aligned} P(\mathbf{x}_{1:n}) &= P(x_1, x_2, \dots, x_n) \\ &= P(x_1)P(x_2 \mid x_1)P(x_3 \mid x_1, x_2) \dots P(x_n \mid x_1, \dots, x_{n-1}) \end{aligned}$$

We haven't yet made any independence assumptions!

Autoregressive Language Modeling

$$\begin{aligned} P(\mathbf{x}_{1:n}) &= P(x_1, x_2, \dots, x_n) \\ &= P(x_1)P(x_2 \mid x_1)P(x_3 \mid x_1, x_2) \dots P(x_n \mid x_1, \dots, x_{n-1}) \end{aligned}$$

$$\begin{aligned} P(\text{the cat sat on the mat}) &= P(\text{the}) * P(\text{cat}|\text{the}) * P(\text{sat}|\text{the cat}) \\ &\quad * P(\text{on}|\text{the cat sat}) * P(\text{the}|\text{the cat sat on}) \\ &\quad * P(\text{mat}|\text{the cat sat on the}) \end{aligned}$$

Important Detail: Modeling Length

- A language model assigns probability to token sequences $\mathbf{x}$.
 - $\mathbf{x}$ can be of any length, and the probabilities should sum to 1 across all possible sequences of different lengths.
- Usually length is modeled by including a “stop symbol” $\langle /s \rangle$ at the end of sequence.
Predicting $\langle /s \rangle$ = modeling stopping probabilities.
 - Relatedly, a “start symbol” $\langle s \rangle$ could also be added to the beginning.

Language model with both start and stop symbols:

$$P(\mathbf{x}_{1:n}) = P(\langle /s \rangle \mid \langle s \rangle, x_1, \dots, x_n) \prod_{i=1}^n P(x_i \mid \langle s \rangle, x_1, \dots, x_{i-1})$$

Why Stopping Probability?

Our language model:

$$P(\mathbf{x}_{1:n}) = P(\langle /s \rangle \mid \langle \mathbf{s} \rangle, x_1, \dots, x_n) \prod_{i=1}^n P(x_i \mid \langle \mathbf{s} \rangle, x_1, \dots, x_{i-1})$$

We need to ensure:

$$\sum_{n=1}^{\infty} \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}) = 1$$

Consider removing stopping probabilities:

$$P(\mathbf{x}_{1:n}) = \prod_{i=1}^n P(x_i \mid \langle \mathbf{s} \rangle, x_1, \dots, x_{i-1})$$

What happens if we don't model stopping probability?

Recall that we need:

$$\sum_{n=1}^{\infty} \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}) = 1$$

Sums of probabilities for all possible length-1 and length-2 sequences:

$$\sum_{n=1}^1 \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}) = \sum_{x \in V} P(x \mid \langle \mathbf{s} \rangle) = 1$$

$$\sum_{n=2}^2 \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}) = \sum_{x \in V} \sum_{x' \in V} P(x \mid \langle \mathbf{s} \rangle, x') P(x' \mid \langle \mathbf{s} \rangle) = 1$$

$$\sum_{n=1}^2 \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}) = 1 + 1 = 2 > 1$$

With the stop symbol...

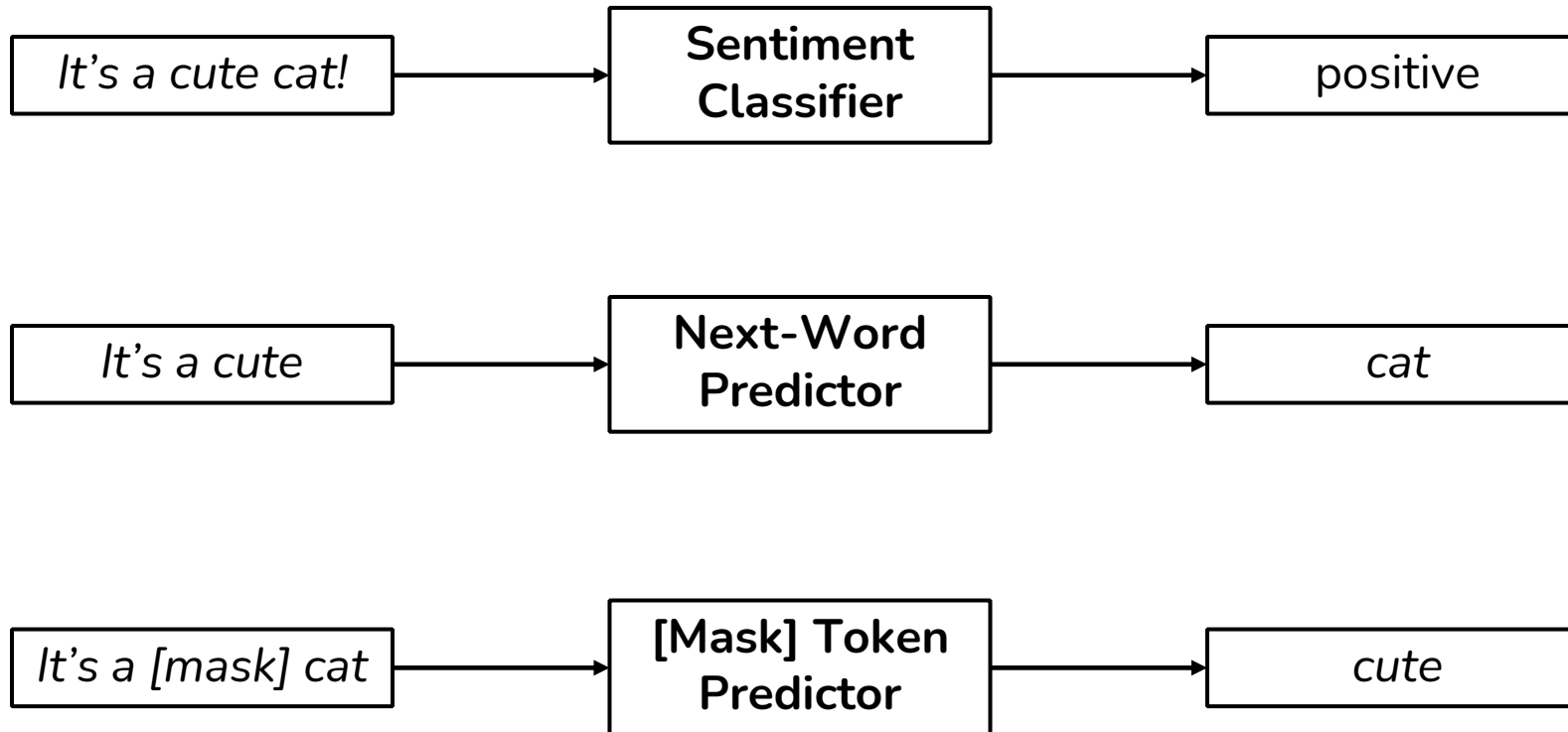
$$\sum_{n=1}^{\infty} \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}, \langle /s \rangle) = \sum_{n=1}^{\infty} \sum_{\mathbf{x}_{1:n}} P(\mathbf{x}_{1:n}) P(\langle /s \rangle \mid \mathbf{x}_{1:n})$$

Proof sketch: once you reach the $\langle /s \rangle$ token after sampling some certain sequence, certain probability mass is “taken away” because $\langle /s \rangle$ is in the same distribution as other vocabulary entries.

Longer sequences that have the same prefix share the “remaining” probability.

Practice: complete the proof that the above quantity is 1 when $n \rightarrow \infty$.

General Formulation: Language Models as Classifiers



Next

- N-gram language models
- Neural autoregressive language models
- Masked language models